

VAPO

Value-Model-Based Augmented Proximal Policy Optimization

State-of-the-Art Reasoning via Value-Model-Based RL
Surpassing DAPO by 10+ Points — No SFT Data Required

60.4

AIME 2024 avg@32

ByteDance Seed · Base Model: Qwen2.5-32B · No SFT Data · 5,000 Training Steps

Why Value Models?

Finer Credit Assignment for Long Chain-of-Thought Reasoning

Value-Model-FREE (GRPO / DAPO)

- Trajectory-level advantage: single reward at end of sequence
- Advantage assigned uniformly across ALL tokens
- Cannot distinguish which step caused success or failure
- High variance Monte Carlo estimates
- Simpler to implement — no value model overhead

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Value-Model-BASED (VAPO)

- **Token-level credit assignment via learned value function**
- Each action's impact on return traced precisely
- **Critical for step-level errors in multi-step reasoning**
- Lower-variance per-token estimates via GAE
- Value model generalises across samples → sample efficiency

Three Challenges Plaguing Value-Model-Based RL

Long Chain-of-Thought Tasks Demand Unique Solutions

① Value Model Bias

Long trajectories combined with bootstrapped learning (Bellman backup) cause value estimates to diverge from true returns.

Error accumulates over hundreds of reasoning steps, making stable training extremely difficult.

② Heterogeneous Sequence Lengths

Responses in long-CoT tasks vary from tens to thousands of tokens.

Short sequences prefer low λ (lower variance); long sequences need high λ (lower bias). A single fixed λ is always sub-optimal.

③ Reward Signal Sparsity

Only the final <eos> token receives a binary reward signal.

In long CoT, the value model must propagate this sparse signal backwards through hundreds of intermediate steps — a severe credit assignment problem.

VAPO Framework: Integrated Solutions

Each Component Targets a Specific Challenge



Length-Adaptive GAE: Dynamic λ Adjustment

Optimising the Bias-Variance Tradeoff Across Heterogeneous Response Lengths

SHORT Response → Low λ

- λ close to 0 (e.g. 0.8)
- Rely more on bootstrapped value estimates
- Prioritises LOWER VARIANCE
- Few tokens → variance dominates error
- Avoids noisy Monte Carlo accumulation

LONG Response → High λ

- λ close to 1 (e.g. 0.98)
- Rely more on accumulated empirical returns
- Prioritises LOWER BIAS
- Many tokens → bootstrapping error compounds
- Avoids value estimation bias across hundreds of steps

Key Insight: A single fixed λ cannot be optimal for sequences of wildly different lengths. Length-Adaptive GAE dynamically tunes $\lambda \propto$ response length → contributes +15 AIME points.

VAPO Scores 60 on AIME 2024

10+ Points Above DAPO and DeepSeek-R1-Zero — Same Base Model, No SFT Data

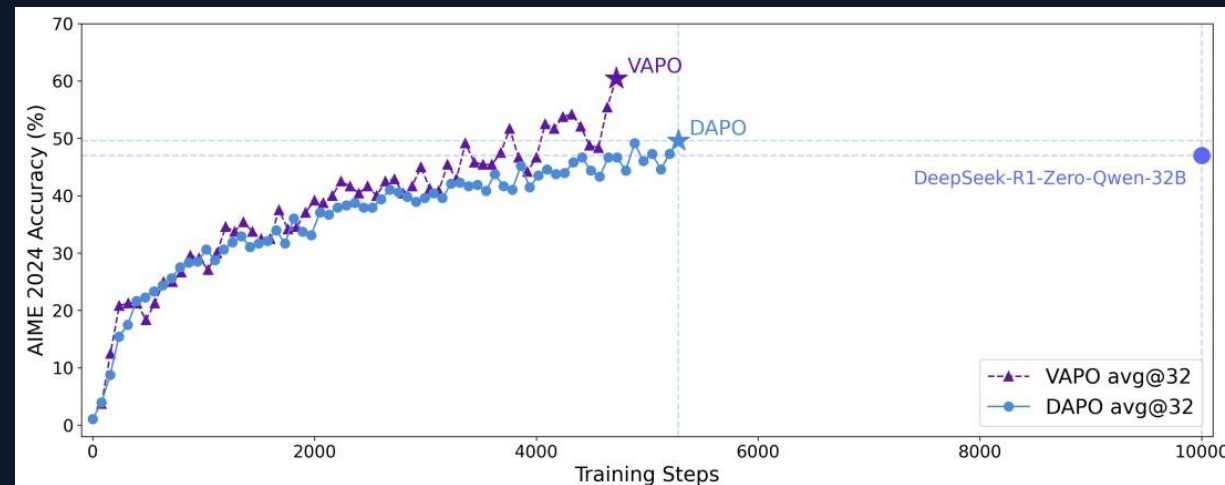


Fig 2: VAPO vs DAPO training curves (AIME 2024 accuracy over steps)

VAPO Reaches SOTA in 5K Steps

Superior Training Efficiency, Stability, and Longer Reasoning Chains

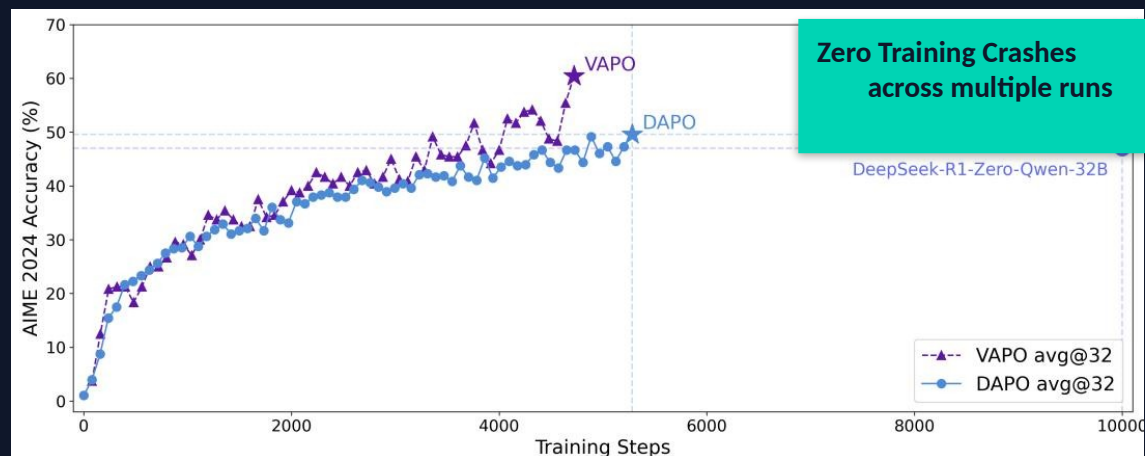


Fig 2: AIME 2024 accuracy over training steps
VAPO reaches 60% at ~4,500 steps; DAPO plateaus at ~50% at 5,500 steps;
DeepSeek-R1-Zero reaches 47% at ~10,000 steps.

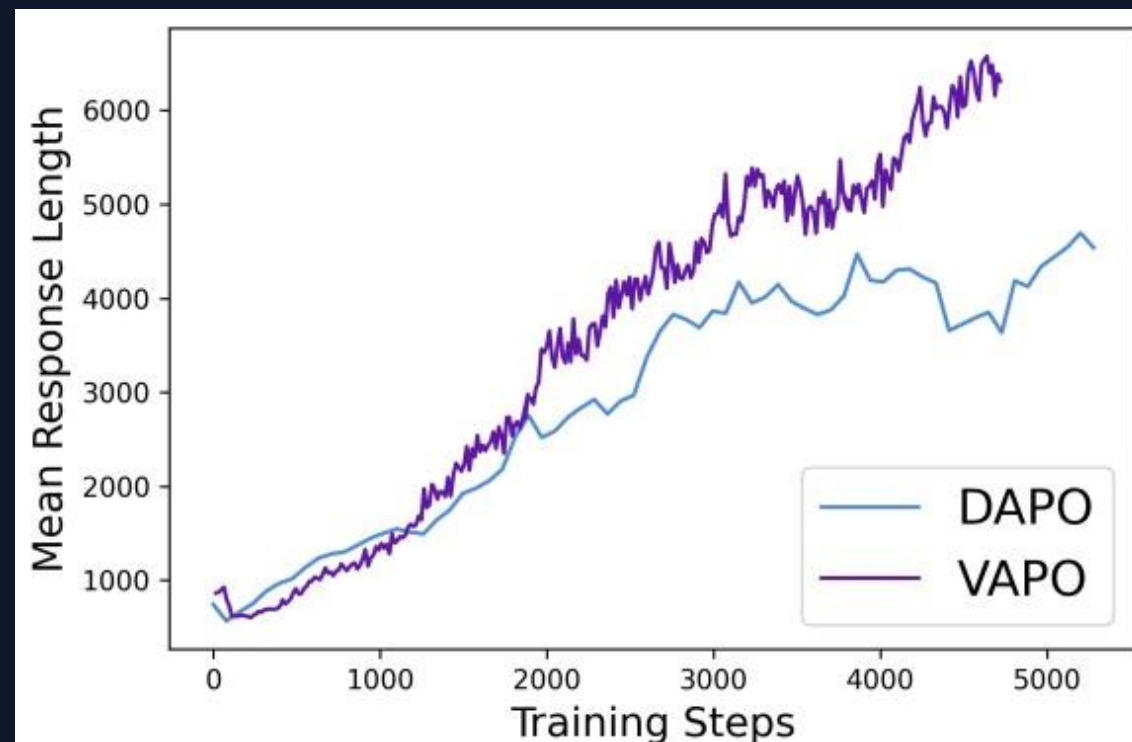
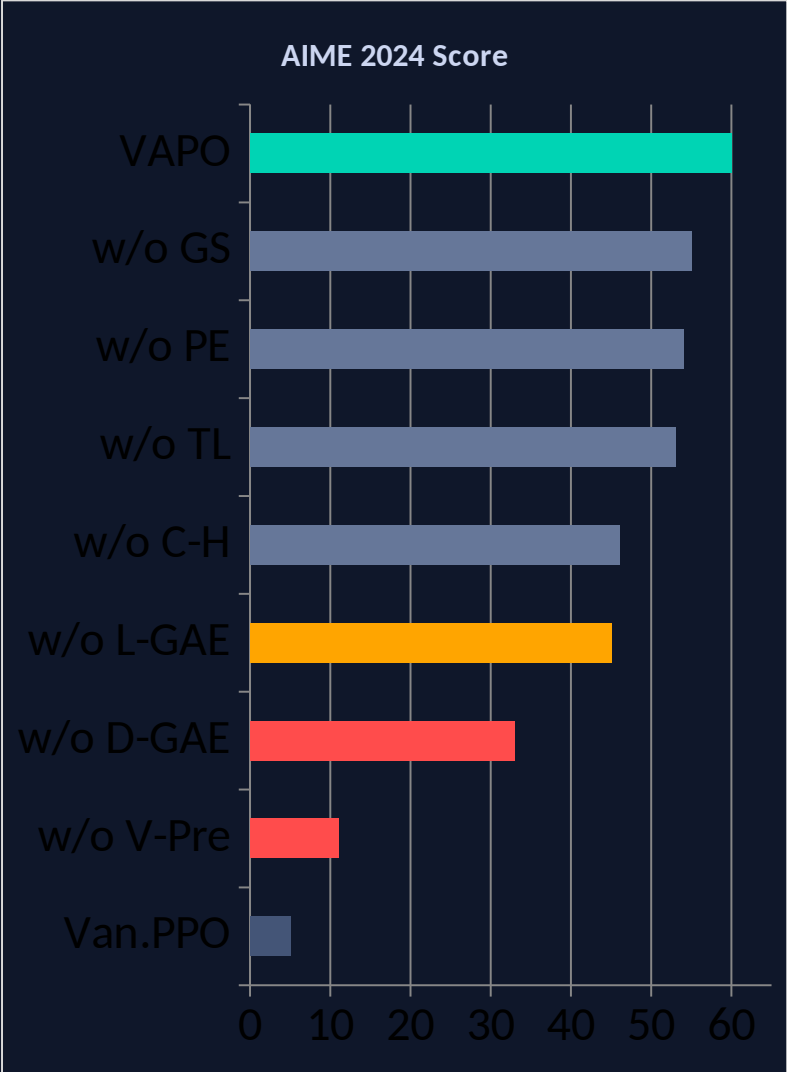


Fig 3: Mean response length over training
VAPO reaches ~6,000 tokens vs DAPO's ~4,500 tokens,
indicating more elaborate reasoning chains.

Ablation: Every Component Matters

Value-Pretraining Alone Contributes 49 Points — Removing It Collapses Performance

Model / Variant	AIME 2024	Δ vs prev
Vanilla PPO	5	
VAPO w/o Value-Pretraining	11	+6
VAPO w/o Decoupled-GAE	33	+22
VAPO w/o Length-adaptive GAE	45	+12
VAPO w/o Clip-Higher	46	+1
VAPO w/o Token-level Loss	53	+7
VAPO w/o Pos. Example LM Loss	54	+1
VAPO w/o Group-Sampling	55	+1
VAPO (Full)	60	+5



Key Takeaways: Value-Model-Based RL Is Back

With the Right Engineering, Value Models Dominate Value-Free Methods

1

Value-Model-Based RL Can Surpass Value-Free Methods

VAPo (60.4) beats DAPO (50) and DeepSeek-R1-Zero (47) on AIME 2024, demonstrating that value models — when properly engineered — unlock superior reasoning performance on the same base model with no SFT data.

2

Value-Pretraining Is the Single Most Critical Component

Removing Value-Pretraining alone collapses AIME score from 60 → 11 (−49 points). Without it, value-model-based RL barely functions at all. Pre-training the value model before RL is non-negotiable.

3

Length-Adaptive GAE Solves the Heterogeneous Sequence Problem

Novel dynamic λ adjustment contributes +15 points. Fixed- λ GAE is provably suboptimal when response lengths range from tens to thousands of tokens — a universal problem in long-CoT reasoning tasks.

4

VAPo Is Stable and Efficient — SOTA in 5,000 Steps

Zero training crashes across multiple independent runs. VAPo reaches state-of-the-art in ~5,000 steps vs DeepSeek-R1-Zero's ~10,000 steps, cutting required training budget roughly in half.

5

Implications: Address Bias, Heterogeneity, and Sparsity Jointly

The three-challenge framework (value bias, heterogeneous lengths, reward sparsity) is a roadmap for future reasoning model development. Ignoring any one challenge severely limits value-model-based RL.